

Inventory Management Module - Best Practice Answers (Industry Standard)

1. Stock Source of Truth

- Single source of truth is Inventory Transactions table.
- All stock should be derived as $SUM(IN - OUT)$.

2. Direct Editing Control

- Users must NOT edit stock directly.
- All changes must occur through transactions (sales, purchase, transfer, write-off).

3. Negative Stock Handling

- Negative stock should NOT be allowed.
- Optional override can be given to admin users.

4. Warehouse / Location Structure

- Items can exist in multiple locations.
- Use structure: Warehouse → Location → Bin (optional).

5. Reservation Logic

- Sales Order reserves stock.
- Delivery/Invoice deducts stock.
- Available = On Hand - Reserved.

6. Cost Calculation

- Cost should be auto-calculated.
- Manual edits after posting should NOT be allowed.
- Supported methods: FIFO, Weighted Average.

7. System Performance

- Use indexed tables and optimized queries.
- Use stock table as cache for performance.

8. Data Issues Prevention

- Prevent duplicate items and transactions.
- Use unique constraints on item code and transaction reference.

9. Transaction Engine

- Use a single centralized transaction table for all movements.

10. Business Priority

- Priority: Accuracy → Auditability → Scalability → Speed.

Inventory Transfer Best Practices

- Stock updates only after posting.
- Cannot transfer more than available stock.
- Partial transfers allowed.
- No editing after posting; use reversal.
- Approval workflow required (Draft → Approved → Posted).

Inventory Write-off Best Practices

- Write-offs require approval.
- Use FIFO or Weighted Average cost.
- Reason is mandatory (damage, expiry, theft, adjustment).
- Tax usually not required.
- Errors corrected using reversal only.

Core Architecture

- Inventory Transactions = single source of truth.
- $\text{Stock} = \text{SUM}(\text{IN}) - \text{SUM}(\text{OUT})$.
- Modules: Purchase(IN), Sales(OUT), Transfer(OUT+IN), Write-off(OUT).

Control Rules

- No direct stock edits.
- No editing or deleting posted transactions.
- Full audit trail required.

Status Workflow

- Draft → Approved → Posted.